Prince Rupert's Problem and Its Generalizations

A Family of Optimization Problems

Terry Ligocki & Greg Huber

In the 1600’s Prince Rupert of the Rhine purposed the following problem:

Given two cubes of the same size, can a hole be cut in one cube such that the other cube can pass through the hole and what is left of the first cube is still connected?

John Wallis found a solution where the hole was centered on the diagonal and the clearance was about 3% at the tightest point.

About 100 years later, Pieter Nieuwland found a “better” solution where the clearance was about 6% at the tightest point.

Here is what Nieuwland's solution looks like:

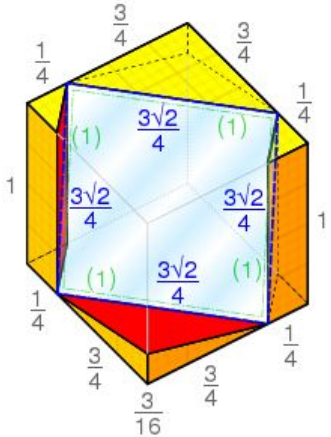
This can be then transformed into an optimization problem:

What is the largest cube that can be passed through a unit cube without disconnecting the remaining pieces of the unit cube?

The answer is one with side length:

$$\sqrt{9/8} \approx 1.06066$$

This has the orientation Nieuwland found and the cross section shown here:



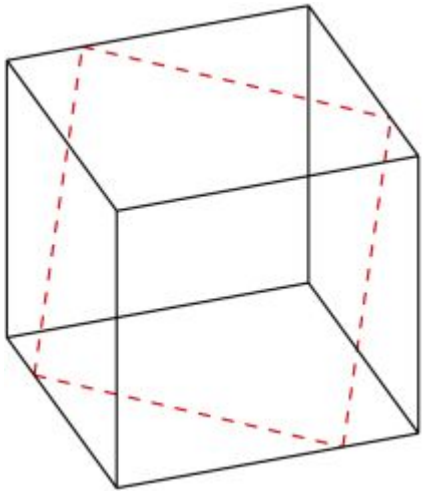
Reformulation and Generalizations

- What is the largest square (2-cube) that fits inside of a unit cube (3-cube)?

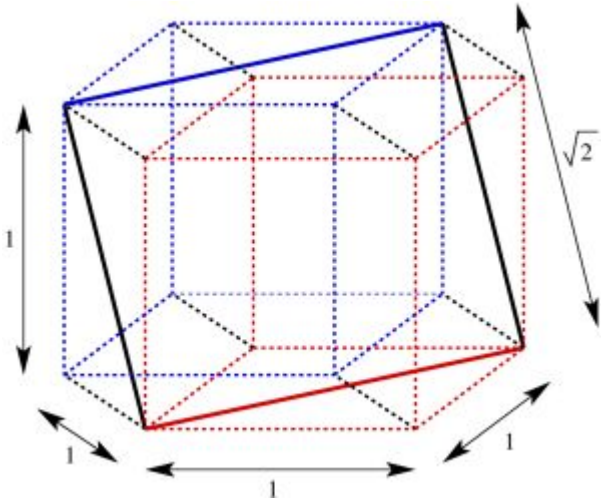
- Generalization: What is the largest m-dimensional cube, m-cube, that fits inside of a unit n-dimensional cube, n-cube?

- For this family of optimization problems, (m,n), only a few solutions have been found and proven:
  - (1,n) - Line segments, 1-cubes, within unit n-cubes
  - (2,3) - The reformulated original problem
  - (3,4) - Posed by Martin Gardner, solved by several readers, and “proven” by Kay R. Pechenick DeVicci Shultz

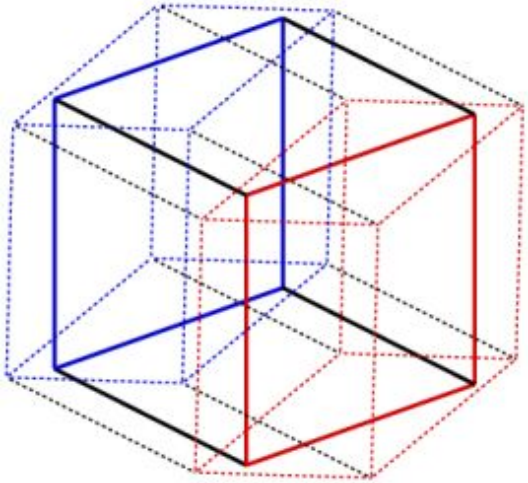
Some Examples



(2,3)



(2,4)



(3,4)

Motivation

- Why are we looking at these problems?

- Why am I presenting these problems to you?
  - Selfish motive
  - Altruistic motive

- Philosophical and historical motivation

- Problems worthy of attack,
  Prove their worth by hitting back
  - Piet Hein

- “Don’t believe everything you think!”

Some Properties and Conjectures

- (1,n) has the solution: $\sqrt{n}$
- (2,3) has the solution: $\sqrt{9/8}$
- (3,4) is the square root of the smallest real root of:

$$4x^4 - 28x^3 - 7x^2 + 16x + 16 = 0$$

- (m,km) = (1,k), (2,2n) = (1,n), (km,kn) = (m,n)
- (2,2k+1) = $\sqrt{k + 1/8}$
- (n-1,n) $\sim 1 + \frac{1}{2}9^{2-n}$

Simulated Annealing

- Wiggle “things” around, wiggle them less each iteration, success!

- Not that simple, the devil is in the details, and the details are unworkable.

- What should we “wiggle”:
  - The corners of what will be an m-cube
  - The m-cube rigidly allowing for changes in size
  - Rotate the m-cube about the center of the n-cube
  - Work directly with a rotation matrix

New Reformulation of the Problem

- Look at all rotation matrices in dimension n, SO(n), this are the n-by-n unitary matrices.

- Start with the m-cube placed axis aligned in first m-dimensions of the n-cube's n-dimensions.

- The optimization problem then becomes:

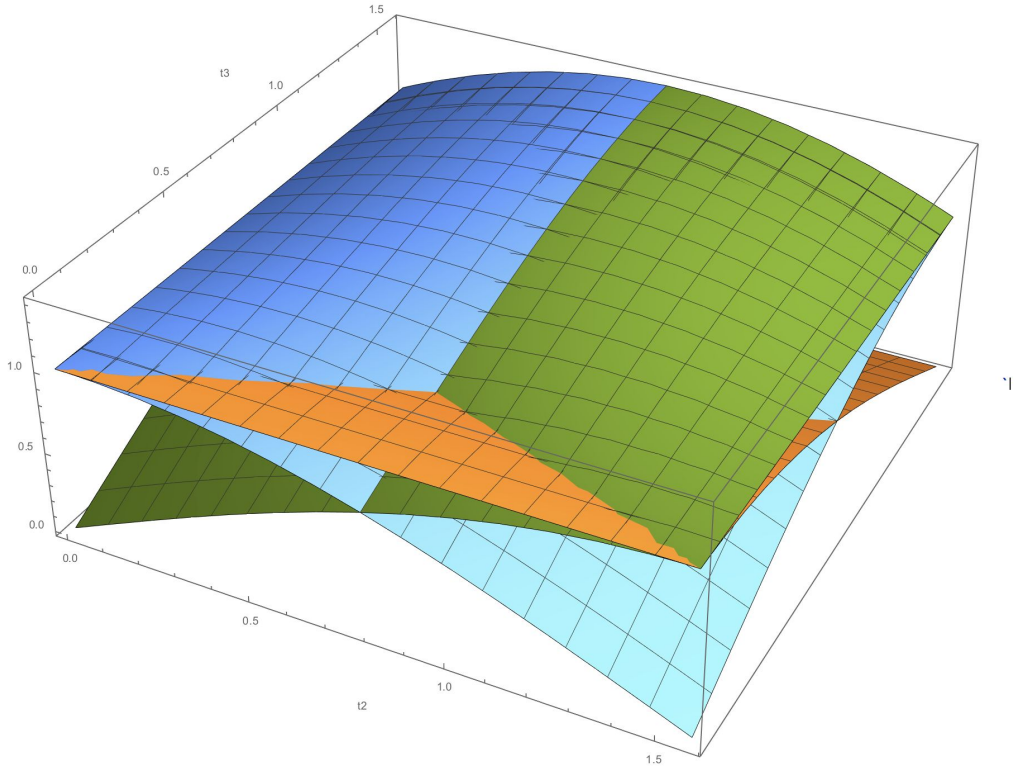
$$\min_{R \in SO(n)} \left(\max_{i=1}^{n} \left(\sum_{j=1}^{m} |r_{ij}| \right) \right), \quad R = (r_{ij}) = \left(\begin{array}{cccc|ccc} r_{11} & r_{12} & \dots & r_{1m} & r_{1,m+1} & \dots & r_{1n} \\ r_{21} & r_{22} & \dots & r_{2m} & r_{2,m+1} & \dots & r_{2n} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ r_{n1} & r_{n2} & \dots & r_{nm} & r_{n,m+1} & \dots & r_{nn} \end{array}\right)$$

Exploring the Space of Solutions

- The space has dimension n(n-1)/2.
- The space can be explored using plane rotations.
- Each step: chose an i,j for the rotation plane and an angle to rotate.
- Run for lots of (m,n) and each cool at different rates.
- Look at the results - both the matrices and the m-cubes.
- Try to discern patterns, make guesses, and learn.

Energy Landscapes

```
In[6]:= Plot3D[{Cos[t3], Cos[t2] + Sin[t2] Sin[t3], Sin[t2] + Cos[t2] Sin[t3]}, {t2, 0, Pi / 2}, {t3, 0, Pi / 2}, AxesLabel → Automatic]
```



What We Learned

- Many of our original conjectures seemed to bear out and the matrices often showed how to relate the their structure to the conjectures.

- All the minimas we have obtained that appear to be global minimas have the vertices of the m-cube on the codimension 2 parts of the n-cube, e.g., the corners of the square lie on the edges of the cube.

- We can explore a space smaller than SO(n) for most of the (m,n) cases because some rotations don't effect the m-cube.

- In the (n-1,n) case only n-1 rotations were needed to get what appears to be the solutions! This was also true for the (1,n) case but that was expected.

What We Learned (continued)

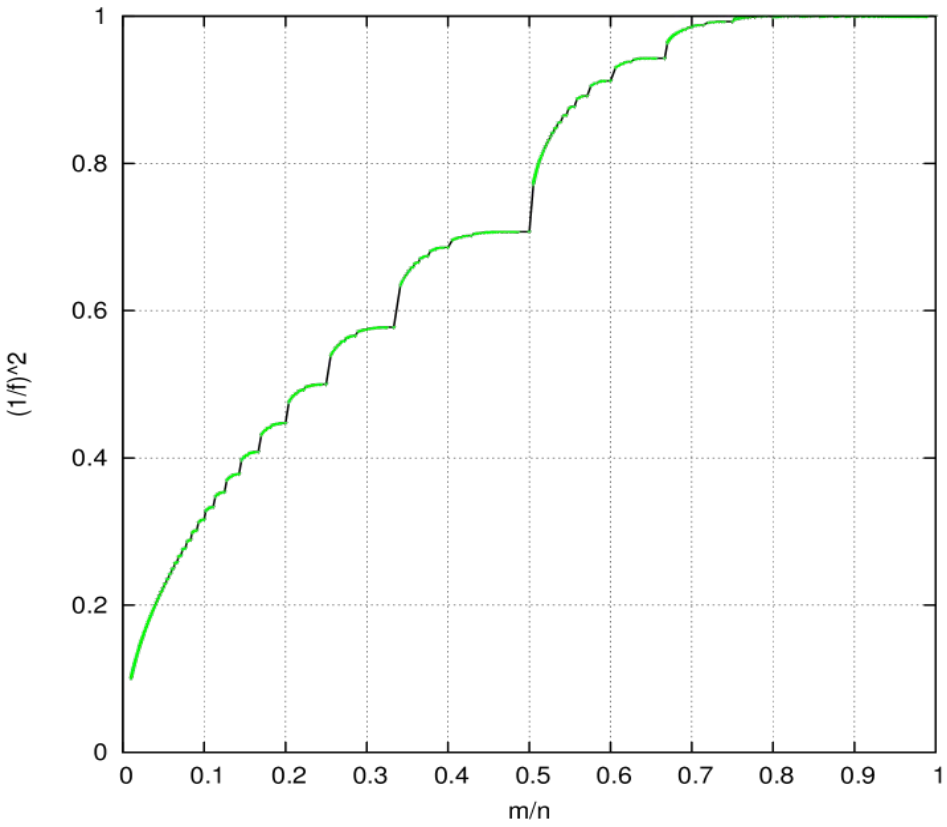
- In all cases, there were a lot more zeros in the matrices than we had expected.
- All the “global” minimas have row sums that are all equal.
- All the “global” minimas seem to occur where various cusps (due to the max and absolute values) coincide.
- Because all the solutions should be the root of some polynomial equation, we could use results that had enough digits to determine these polynomials.

Some Matrix Structures

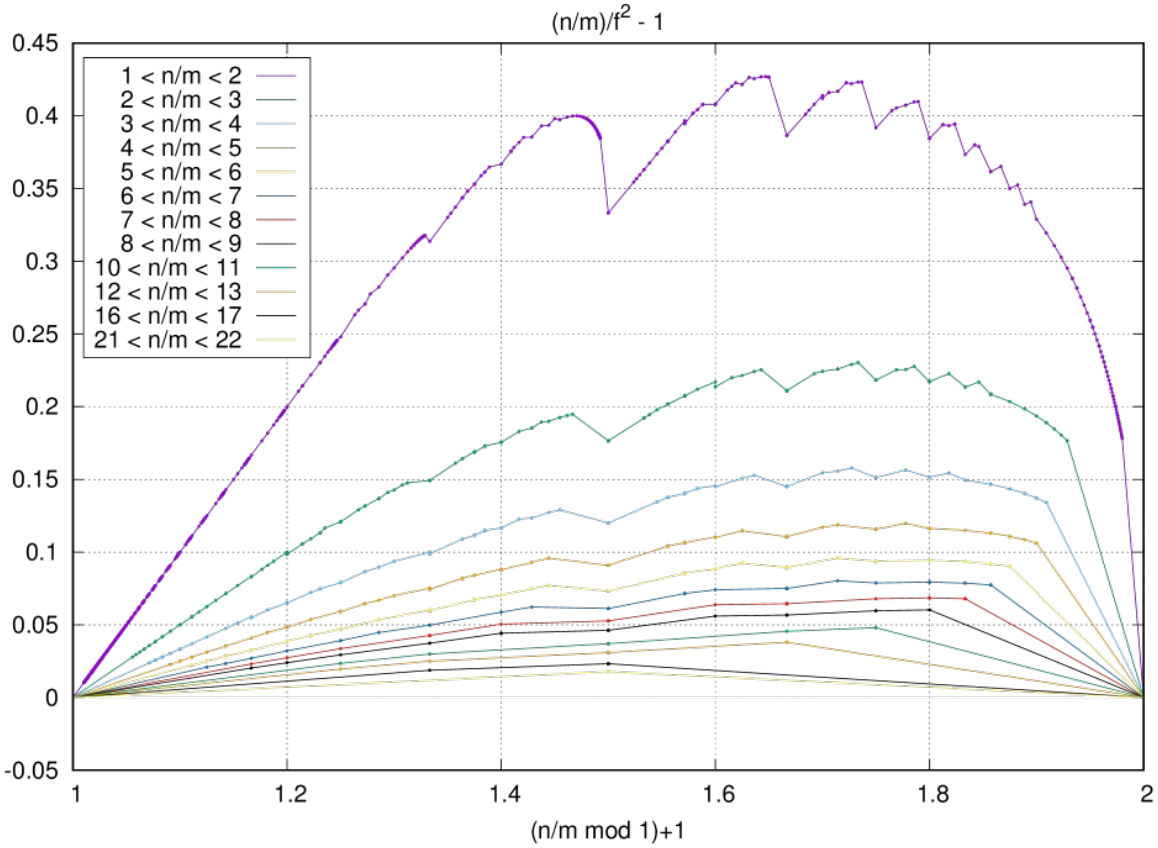
$$R_{km,kn} = \begin{pmatrix} \overbrace{R_{m,n} \quad 0 \quad \dots \quad 0}^{\text{k times}} & \Big| & \dots \\ 0 \quad R_{m,n} \quad \dots \quad 0 & \Big| & \dots \\ \vdots \quad \vdots \quad \quad \vdots & \Big| & \vdots \\ 0 \quad 0 \quad \dots \quad R_{m,n} & \Big| & \dots \end{pmatrix}$$

$$R_{n-1,n} = \begin{pmatrix} r_{1,1} & r_{1,2} & \dots & r_{1,n-1} & \Big| & r_{1,n} \\ r_{2,1} & r_{2,2} & \dots & r_{2,n-1} & \Big| & r_{2,n} \\ 0 & r_{3,2} & \dots & r_{3,n-1} & \Big| & r_{3,n} \\ \vdots & \ddots & \ddots & \vdots & \Big| & \vdots \\ 0 & \dots & 0 & r_{n,n-1} & \Big| & r_{n,n} \end{pmatrix}$$

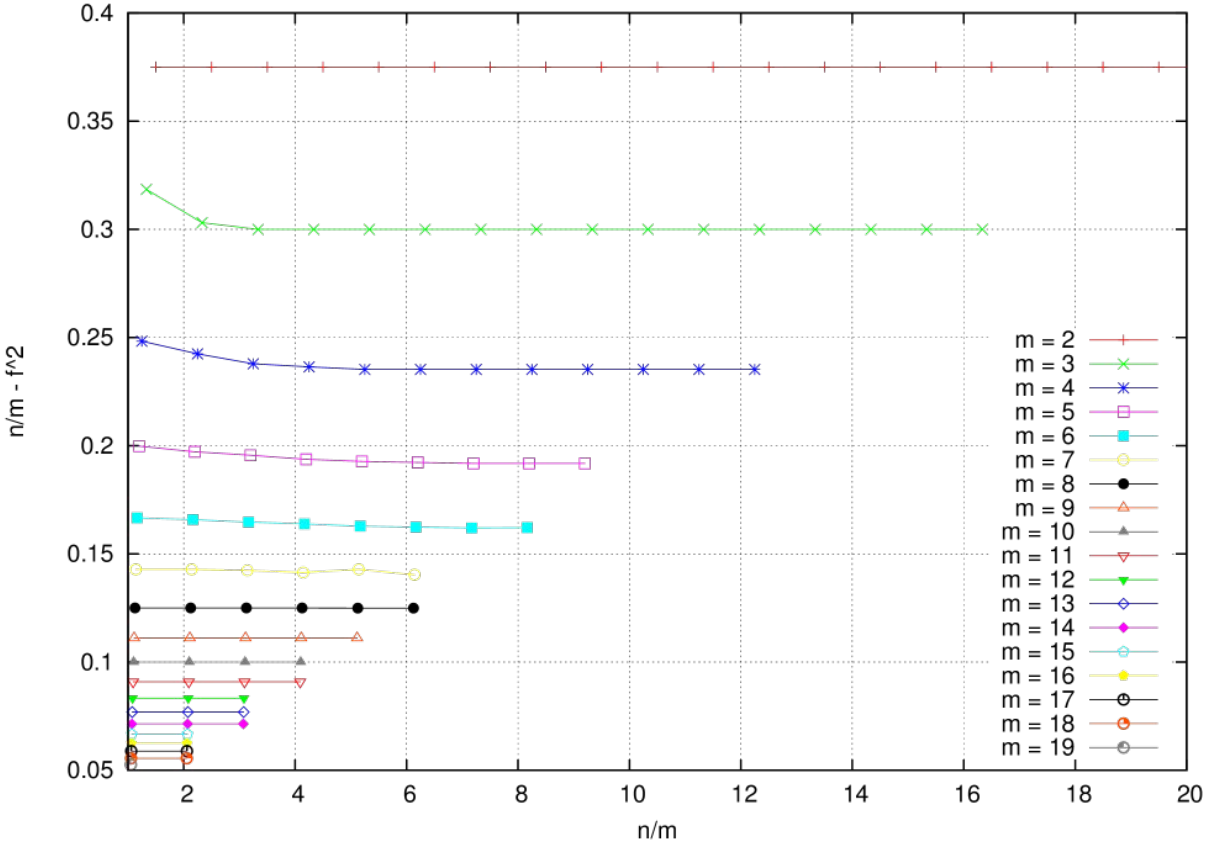
Graphs of Results - “Fractal” Structure



Graphs of Results - Overlaying Structure



Graphs of Results - Families



New Methods Based on Simulated Annealing Results

- For the (n-1,n) case we developed a fixed point iteration that used the (n-2,n-1) solution to generate an initial guess and converged quickly. Using Mathematica and a large number of digits, we were able to get results up to n = 100. This technique may work for other cases, e.g., (n-2,n).

- Based on the “families” and their matrix representations we were able to find a way to extend lower dimensional solutions to higher dimensions possible solutions and/or starting points for other iterations or searches.

- Some method of making the row sums equal and accounting for all the zeros may lead to enough constraints to solve for the global minimas.

Nagging Questions

- How should we be representing SO(n)? Currently we:
  - Represent an element of SO(n) by the product of n(n-1)/2 plane rotations is a specific order. This representation isn't unique and it doesn't reflect the manifold structure of SO(n).
  - Move around SO(n) using plane rotations. Again, this doesn't relate to the coordinate systems of a set of charts of SO(n).

- If we were to represent SO(n) differently, would many of the issues with local minima and other minimization techniques be less severe?

- Can we "soften" the max and absolute value functions to get a smoother energy landscape, utilize this, and still solve the original problems.